R13

Code No: 126ER

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
SOFTWARE TESTING METHODOLOGIES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What is a dichotomy in software testing?	[2]
b)	What are the differences between a functional bug and a non-functional bug?	[3]
c)	Explain the concept of dataflow testing and its significance in software testing.	[2]
d)	List and explain the different possible interpretations of decision symbols.	[3]
e)	What is a domain in the context of software testing?	[2]
f)	How does the design and architecture of a software system impact its testability?	[3]
g)	What is a path expression in software testing?	[2]
h)	What are the main categories typically used in a KV chart for software testing?	[3]
i)	Define the power of a matrix.	[2]
j)	What are some common challenges faced during state testing, and how can they be overcome?	[3]

PART - B

(50 Marks)

2. Discuss the challenges of achieving complete coverage using path predicates in software testing. How can these challenges be addressed? [10]

OR

3.a) What are some common techniques to derive path predicates from a control flow graph? Explain.
b) How does path sensitizing help in detecting complex program behaviors or corner cases? [5+5]

4.a) What are the main components of a transaction flow graph, and how do they relate to each other?
b) Explain the concept of dataflow coverage criteria such as All-Defs, All-Uses, All-DU-Pairs, and All-Paths. How do these criteria help in measuring the effectiveness of dataflow testing? [5+5]

OR

5.a) What are the main challenges faced when performing dataflow testing, and how can they be addressed?
b) How can equivalence partitioning be applied to transaction flow testing? Provide an example to illustrate your answer. [5+5]

6.a) Distinguish between a feasible path and an infeasible path.

b) Discuss the characteristics of a Nice domain and why it is considered desirable for testing. [5+5]

OR

7.a) How can test case prioritization techniques be applied to Nice and Ugly domains to optimize testing efforts?

b) Why is it important to test values at the boundaries of domains? Explain. [5+5]

8. Consider a decision table with multiple conditions and actions. Describe the process of identifying and eliminating any redundant conditions or actions from the decision table. [10]

OR

9.a) How can a KV chart assist in prioritizing testing efforts?

b) Explain the purpose of a reduction procedure in path expressions. [5+5]

10.a) Explain the node reduction algorithm and its role in optimizing state graphs or finite-state machines.

b) What is state testing, and why is it important in software development? [5+5]

OR

11.a) Describe the criteria or metrics used in the node reduction algorithm to determine which nodes can be merged or eliminated.

b) Describe the process of representing a relation as a matrix. [5+5]

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